Empirical Evaluation of Chronomem: Recall Retention, Compression Behaviour, and the Limits of Recursive Summarization

YSTYLE[1] · ORCID: 0009-0009-6929-0070 · lxy5266@live.com
DeepSeek V4.1 Flash[2] · AI co-author

[1] *Empirical companion to the position paper (`arXiv`/AiraXiv 2610.0035). Implementation, evaluation harness and raw result archives:* `https://atomgit.com/ystyle/chronomem`. [2] *AI co-author, as credited on the AiraXiv preprint. See the AI-use disclosure at the end.*

Abstract

The position paper for Chronomem (忆时塔) specified an evaluation plan and reported no measurements. This paper carries it out. We evaluate a time-layered external memory service — which separates **exact recall** (reference-chain navigation to raw records, no LLM call) from **semantic recall** (reading LLM-generated window summaries) — along three axes: how recall degrades with the age of the fact, how much information the recursive summarization chain preserves, and what the system actually retained over two months of real use.

Three findings. **(1) The recency baseline's failure is structural, not gradual.** A bounded recency window answers perfectly at a one-day lag and then drops to zero, and an ablation over six window sizes shows the cliff tracks capacity (20 items → 1 day; 200 items → 7 days). One-shot compaction's score depends entirely on what it preserves: a variant that keeps each item's value answers 100%, while one that paraphrases values away scores zero at every lag including the most recent — a 100-point gap produced by 11 tokens per question (196 vs 185). The load-bearing thing in compression is the item-to-value *relation*. Layered exact recall holds 100% across the full 14-day span, with zero variance across seeds. A coarse-summary arm, added after we noticed the original comparison never exercised the layer the architecture is named for, shows the coarse level is not a cheaper version of the fine one: it reads *more* text per question (195 vs 88 client tokens) and answers correctly about a third as often (18.8% vs 53.1%), because compressing six days into one summary drops the pairing between an item and its value while keeping the item's name.

(2) The recursive step's fidelity depends on input diversity, not on the level. Measuring the step correctly — same project, same 7-day window — shows that with a single day of input the step usually declines to compress: 68% of such pairs come back at ≥0.95x length, many of them character-identical to the input, giving 85% token retention at a 0.90x ratio overall (though 19% do compress substantially). With two or more days of input the step actually compresses (0.53x) and retention falls to 58% for quantities, 48% for causal connectives. Aggregated naively, the single-day cases inflate fidelity from 58% to 81%.

(3) Over two months of real use, what survives is names, not numbers. A blind recall test — an agent allowed to consult only the memory service, never the repository — answered 6/6 prepared questions correctly, all via the no-LLM path, including exact figures. But along the summary chain the first casualties are quantitative evidence and causal mechanism, while proper nouns and identifiers persist longest. We also report two measurement traps that

produced confidently wrong results before being caught: a length artefact that made the coarser level look *more* faithful, and a grouping error that made a single day's input dominate the average.

Finally, we report a negative result about evaluation methodology itself: independent LLM judges produced specific claims that did not survive verification — two of the two claims we checked in detail were wrong, both in the direction of overstating loss. Conclusions here therefore rest on mechanical, reproducible measurements, with judge output used only to indicate where to look.

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1. Introduction

The position paper introduced Chronomem as a time-layered external memory service and stated its central design claim: recall is not one operation. **Exact recall** walks materialized reference chains down to raw records and calls no LLM; **semantic recall** reads LLM-generated summaries and is the only form that genuinely compresses coarse windows. It closed with an evaluation plan and no results.

This paper reports those results. The plan had three parts, and we address each:

- **(i)** a controlled comparison against flat memory and one-shot compaction, on recall accuracy and token cost;
- **(ii)** a semantic-drift study along the summary chain;
- **(iii)** a study of the work-log use case on the authors' own continuous deployment.

Two things distinguish this evaluation from a synthetic benchmark exercise. First, part (iii) runs on a deployment that has been accumulating real records for roughly two months, so the corpus is not invented. Second, in carrying out (ii) we found that the *obvious* way to measure it is wrong in a way that inverts the conclusion, and we report the correction in detail — the measurement method is part of the result.

We also report a methodological negative result (§6). We ran independent LLM judges over the same material; of the two specific claims we verified in depth, **both were wrong**, both overstating information loss. This is not a claim that judges are useless — they pointed at two real phenomena worth investigating — but it does mean their counts cannot be cited, and it motivates our reliance on mechanical measurement throughout.

1.1 Summary of contributions

1. A controlled comparison over five arms — recency, one-shot compaction, exact drill-down, day-level summaries and **week-level summaries** — showing that the recency baseline's degradation is a **capacity cliff** rather than gradual decay, verified by a six-point ablation (§3).
2. A corrected measurement of the **recursive summarization step**, showing that its fidelity is governed by *input diversity*: single-input steps usually restate their input rather than compress it (68% at ≥0.95x, 19% compressing substantially), while multi-day steps halve the text and lose ~40% of quantities (§4).
3. A characterisation of **what survives** along the chain on real data: identifiers outlast causal connectives, which outlast quantitative evidence (§5).
4. Evidence that the no-LLM path is **sufficient for exact recall** of concrete operational facts months later, via a blind recall test scored against independently established ground truth (§5.1).
5. A **negative methodological result** on LLM judges, and two measurement traps documented with their corrections (§6).

2. Method

2.1 Two corpora, deliberately different

Synthetic trace. A seeded generator (`bench/gen_trace.py`) emits N simulated days of agent activity: each day produces a small number of *facts* (a decision with a recorded outcome) plus filler activity, and a probe set asks about facts from earlier days. Constraints established by measurement during development:

- **Answers are assigned round-robin**, not uniformly: a uniform draw over a small probe set clusters, so with 9 probes one option took 7 (a 78% majority-class baseline that leaves no headroom to detect a real difference). Round-robin bounds that baseline near 1/|choices|.
- **Topics are drawn without replacement per day.** Repeating a topic within a day makes "what was chosen for X on day N" ambiguous; 6 of 54 (day, topic) pairs were ambiguous, and those accounted for *every* miss the layered backend showed — the system had retained the fact and the extractor picked the other one.
- **Probes always look backwards, and the lag grows with the run.** Same-day probes are answered by any recency window (a 20-item window scored 7/9); a fixed short lag is likewise answered by any bounded window (a 20-item window still scored 90.6% on a 30-day trace with a 1-day lag). The lag therefore grows with the day, which is the axis a layered store should win on.
- Each fact carries an **opaque reference token** (`ref ZQ01`) inside the text. Summaries reproduce it verbatim, which makes extraction unambiguous — keying on (topic, day) is unreliable because a summary's prose conclusion routinely names several topics and several options in one sentence.

Deployment corpus. The authors' own Chronomem deployment, running continuously since 2026-08-29, single-user. At evaluation time it held 480 work-log records, 1201 agent-activity records and 63 lesson records, with 37 day-level and 6 week-level summary windows for the work log. This corpus has no ground truth, so it is used for *survival* measurements (§4, §5) rather than accuracy.

2.2 Five backends under identical input

Backend	Mechanism
`flat-recency`	keeps everything, exposes the most recent 20 entries
`one-shot-compaction`	rewrites history into a running report every 2 days, discarding detail
`chronomem-exact`	reference-chain drill-down to raw entries (no LLM call)
`chronomem-day`	reads the **day-level** LLM summary of the fact's own window
`chronomem-week`	reads the **week-level** LLM summary containing the fact

All arms see the same trace, answer the same probes, and are read by the **same extractor** — a weak extractor depresses all of them equally and leaves the comparison intact, whereas a per-backend reader would confound the two.

A missing arm, and why it matters. Our first version of this comparison had no coarse-summary arm: the summary backend read the *day-level* summary only. For a paper whose claim is about *layering*, that is a real gap — the "layered" side of the comparison exercised exactly one layer, and the coarse level — the thing that is supposed to

trade information for span — was never asked a question. We therefore added a fifth arm, `chronomem-week`, which answers each probe from the week-level summary containing the fact.

The two summary arms are not two granularities of the same measurement. A day arm reads the window the fact came from, so the question and the material have the same extent; a week arm reads a window covering six other days, so the answer is one compressed item inside a larger bundle. That asymmetry is precisely the trade the architecture claims, and it is only visible if both arms are present.

The runner ingests the whole trace before asking anything (so that summary-generating backends can do off-line work), which means bounded backends must filter by the day being asked about. Without that filter the recency baseline answers day 5's question from day 30's most recent events and scores as well as a system that actually remembers.

2.3 What "lag" means, and why buckets are ranges

Probes are bucketed by lag (days between the fact and the moment of asking). Bucket edges are 0/1/2/3/5/7/14/30/60, so a bucket labelled 7 holds lags 7–13, and on a 30-day trace the top bucket contains 29-day probes. Tables report the true span (`7-13`, `14-29`): reporting the lower edge as if it were the lag invited a capacity conclusion the data did not support.

2.4 Measuring the chain: three attempts, two wrong

This is worth stating explicitly because the wrong versions produced confident, publishable-looking numbers.

Attempt 1 — whole day vs whole week. Retention came out at 81–84%, suggesting the coarse level was nearly lossless. It is not a valid comparison: a week bundle covers 5–7 days and runs ~3.4x longer than a day bundle, so it contains the measured day's content *plus* six other days'. Bigger target text contains a given token more often for no reason. In a variant that compared the full day against the full week, the artefact was strong enough to make the coarser level look *more* faithful than the finer one (42% vs 29% for quantities) — a result that should have been immediately suspect.

Attempt 2 — one day vs the week's matching project section. Still unequal: the week section covers the whole week.

Attempt 3 (used). Restrict *both* sides to the same project and the same 7-day window — which is what the recursion actually transforms, since the summarizer aggregates a project's child summaries by group key. Only then does the effect of interest appear (§4).

2.5 Mechanical token survival

To avoid judging prose, survival is measured over **tokens classified by kind**:

- **numeric** — quantities of 2+ digits, ratios, decimals (years excluded as labels);
- **symbolic** — backticked identifiers, `--flags`, camelCase and snake_case names;
- **causal** — connectives marking reasoning (因为/由于/根因/导致/因此/所以/退化成/…).

For each level we report the fraction of the finer level's tokens still present. This is objective and repeatable. It does not judge whether a human would say a fact "survived"; it measures whether the specific carriers of information did.

`absent` (the level exists but omits the fact) and `missing` (no summary exists for that window) are counted separately throughout. Conflating them reports non-generation as information loss — a mistake made twice during development, once here and once in an earlier drift study.

We attempted the complementary semantic metric and report that it failed. The reviewer's objection to token counting is that it confuses lexical with semantic preservation: a fact restated in different words scores as lost, and a fact whose tokens remain but whose meaning is inverted scores as kept. The natural fix is an embedding-based measure — similarity of a level's text to the fact, against a control fact from another window — and `semantic_survival.py` implements exactly that. It does not work here, and the reason is worth stating rather than hiding:

probe	cosine similarity
identical texts	1.0000
semantically close	0.9772
topically related	0.9901
completely unrelated ("time-series database stores timestamped data" vs "I had tomatoes and eggs for lunch")	**0.9758**

Unrelated texts score 0.9758 while close ones score 0.9772 — a range of 0.0014 across the entire semantic spectrum, against 1.0 for identity. Consequently the margin of a fact against its own window and against a control window is ~0 for every fact measured (+0.001 at day level, +0.002 at week level), and the metric can neither support nor refute anything.

Diagnosis: the embedding server runs with `--pooling cls`, whereas Qwen3-Embedding requires `--pooling last`; applying the model's documented query-instruction prefix made separation *worse* (identical texts fell to 0.798), which is consistent with a pooling misconfiguration rather than a misuse. We did not change the endpoint's configuration — it is shared infrastructure, and re-tuning it to make one paper's metric work would mean measuring a different system than the one that runs.

What this does and does not mean for the results. It does not mean the token metric is validated; its stated limitation stands unaddressed. It does mean the limitation cannot be closed by the obvious route on this deployment, and that a semantic replication needs a correctly configured embedding model. We note one asymmetry from the failed attempt that is at least suggestive: of 17 facts, 11 were scored kept by the token metric and lost by the semantic one, and none the other way — the direction the reviewer predicted, though on a metric we have just shown to be uninformative, so we do not treat it as evidence.

2.6 Summarizer configuration

Reported because the reviewer asked for it explicitly, and because the variance above is only interpretable against it:

setting	value
Model	`qwen3.5-4b` (Q4_K_M GGUF, served by llama.cpp behind an OpenAI-compatible endpoint)
Sampling temperature	**0.7 — the endpoint's default; the client sends no temperature**
Seed	not sent (endpoint default)
Context (`--ctx-size`)	131 072 (`CHRONOLOG_SUMMARY_CONTEXT_TOKENS`)
Output cap	4 096 tokens (`CHRONOLOG_SUMMARY_MAX_TOKENS`)
Reasoning	disabled (`chat_template_kwargs.enable_thinking=false` , `CHRONOLOG_SUMMARY_NO_THINK`)
Per-call input cap	40 000 tokens; up to 8 calls per job
Provider	`openai`

The client does not request deterministic decoding, so every summarization is a fresh sample from a 0.7-temperature distribution. We did not attempt to lower it: the deployment's summaries are generated by a scheduler that runs unattended, and freezing the temperature would make the reported retention numbers describe a configuration the deployment does not use. The variance is therefore reported as a property of the system as operated, and it is the reason the day/week arms are read as distributions rather than point estimates.

2.7 Isolation

The synthetic experiments write, so their client **refuses** the production port at construction. The real-corpus readers default *to* production — the real corpus exists nowhere else — and therefore allow only read-only tools, enforced in code rather than documented. During development a default pointing at production caused a synthetic trace to be written into the live store; it was removed and the guard added.

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3. Recall retention: the baseline fails structurally

3.1 Setup

30-day synthetic trace, 300 events, 32 probes, three seeds (7/11/23). Accuracy per lag bucket, averaged over seeds; `±` is the standard deviation across seeds. Figure 1 plots the shape; the tables give the numbers.

3.2 Results

lag (days)	flat-recency	one-shot compaction (destructive)	one-shot compaction (conservative)	chronomem exact	chronomem day	chronomem week
1	100%	0%	100%	**100%**	100%	0%
2	0%	0%	100%	**100%**	50%	0%
3–4	0%	0%	100%	**100%**	25%	0%
5–6	0%	0%	100%	**100%**	25%	0%
7–13	0%	0%	100%	**100%**	57.1%	28.6%
14–29	0%	0%	100%	**100%**	66.7%	33.3%
overall	6.2%	**0.0%**	**100.0%**	**100.0%**	**53.1%**	**18.8%**

Exact recall is 100% in every bucket, as before. The two summary arms are single runs (one seed) and their per-bucket counts are 2–14 probes, so the bucket-level figures are noisy; the totals are the more reliable reading.

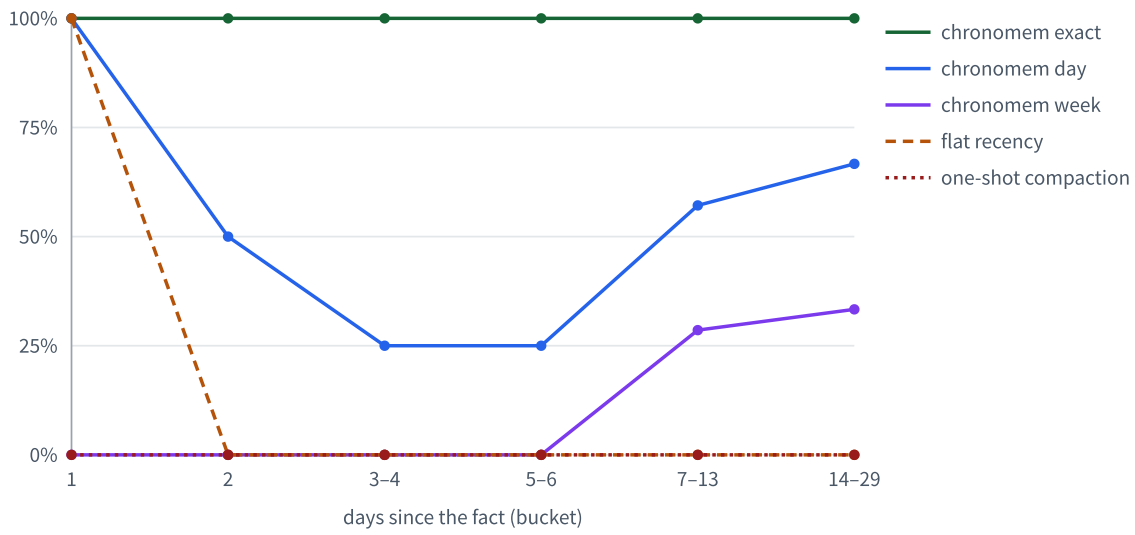


Figure 1. Recall retention by lag, five arms (seed 7). Exact recall is flat at 100% — it does not decay with age. The recency baseline is perfect at one day and zero from two days on, a cliff rather than a slope. One-shot compaction is zero everywhere, including the most recent window. The two summary arms lose information but not monotonically, because a summary is written once and does not change as it ages (§ 3.2c). Dashed styles distinguish the arms if colour is lost.

(d) The coarse level is not a cheaper version of the fine one — it is a different trade. The intuitive story would be "read the week summary, get a shorter answer". The measurement says otherwise:

arm	accuracy	client tokens/probe	server tokens/probe
`chronomem day`	53.1%	**88**	37 500
`chronomem week`	18.8%	**195**	43 750

The week arm reads *more* text per question, not less, and answers correctly about a third as often. The reason is visible in a single response. Asked which option `ZR01` took on day 1, the day summary says `ZR01 (bob): 选择方案`

`A`; the week summary says:

> Day 1 至 Day 6 期间，针对 ZT02、ZR01、ZV04、ZU03… 等项目的 … 模块，依次选定 Option A/B/C

Ten reference tokens, one sentence, and **the mapping from token to option is gone**. The week text is longer than the day text (764 vs 343 characters) because it has to mention six days' worth of identifiers — but it mentions them as a list, not as answers.

So the coarse level's cost profile is not "smaller output": it is a **summary of summaries whose compression drops the pairing between an item and its value**, while retaining the items' names. That is consistent with §5.2's finding on real data — identifiers outlive the quantities and mechanisms attached to them — and it means a client that needs a specific fact from a month ago is not simply paying less for a coarser answer; it is paying a similar amount for an answer that may not contain the fact at all.

The layered design is still what makes recall over long spans possible — exact recall is 100% at every lag and the day level retains over half — but the gain does not come from the coarse summary being cheap. It comes from the fine levels remaining available beneath it, and beyond the coarsest level from paging the raw series directly (§7).

This is the same mechanism §4 characterises from the other direction. There, a week summary built from a *single* day's summary usually restated its input (no compression, no loss); here, a week summary built from six days actively merged them and lost the item-to-value pairing. **The coarse level's fidelity is a function of how much its input varies, and its cost is a function of how much it has to name** — neither follows from the level's position in the hierarchy.

Three observations, in decreasing order of how much they surprise us.

(a) The recency baseline's drop is a cliff, not a decay. It is perfect at lag 1 and zero from lag 2 onward. The window holds 20 items and the trace emits ~10 events/day, so the boundary falls at ~2 days. Nothing about the *content* changes across that boundary — the information is simply absent from the target text.

(b) One-shot compaction scores zero everywhere, including lag 1 — and that zero is a property of the compressor, not of compaction. This was the sharpest objection to our first draft, and it is correct as stated: we wrote the compactor to paraphrase values away, so its zero is partly our doing. To separate the two, we added a **conservative compaction arm** that compacts on the same schedule but preserves every token-to-value pair, dropping only context and prose:

compaction variant	accuracy	client tokens/probe	server tokens/probe
destructive (rewrites values into paraphrase)	**0.0%**	185	120
conservative (keeps token→value pairs)	**100.0%**	**196**	120

Two things follow, and both matter. First, the objection stands: a compactor that keeps the pair loses nothing here, so "compaction cannot answer" is too strong as a general claim. Second, the two variants differ by **11 client tokens per probe (196 vs 185)** — under 6% — while differing by 100 points of accuracy. The failure is therefore not caused by compaction being aggressive; it is caused by the compression **dropping the pairing between an item and its value**, which is the one thing a probe asks for. That is the same mechanism §4 finds in real summaries (identifiers survive, the values attached to them do not) and it sharpens rather than weakens the structural claim: what a compressor must preserve is the *relation*, and paraphrasing is what destroys it.

We report the conservative arm in the main comparison from here on.

(c) The layered store's loss is compression loss, not temporal decay. Exact recall holds 100% at every lag with **zero variance across seeds** (all six buckets, std 0.0). Summary recall does not decline monotonically with lag — 78% at 14–29 days is *higher* than 55% at 7–13 days — because a summary is produced once and does not change as it ages. What it loses is what compression dropped at generation time; time is not the variable.

The variance column is informative, and we now quantify it. Exact recall is deterministic (std 0.0 in every bucket, since the reader resolves a reference chain). Summary recall's spread comes from **LLM sampling**, and it is large. Taking two real production windows, importing their raw entries into the bench instance and re-summarizing each five times with `recompute` (identical input, identical parameters, `temperature` left at the endpoint default of 0.7):

source window	input	five output lengths	length SD	mean pairwise similarity
2026-09-15	12 entries / 13 724 chars	1789, 1686, 1332, 1205, 1258	**265**	**0.263**
2026-09-12	33 entries / 29 198 chars	2350, 1562, 1456, 1432, 2189	**437**	**0.250**

Two runs of the same window agree on only about a quarter of their text. The differences are substantive, not cosmetic — the same day is reported once as "权限为 `admin-only`" and once as "团队权限链路重构（A1/A2）". So summary bucket means are centres of a wide distribution rather than precise values, single-run bucket figures are indicative only, and the day/week comparison in this section carries that spread.

A correction worth recording. An earlier draft of this section reported "237 vs 453 characters" as evidence of sampling variance. That number came from reading `mem/recall` after two `recompute` calls and comparing the two replies — but `recall` returns *all* stored versions concatenated, newest first, so the second reply was the first version plus a second one, not a second sample. The figure measured accumulation, not variance. The numbers above come from re-running properly and reading one version at a time.

3.2b Statistical power: which claims the sample size can carry

Bucket accuracies rest on 2–14 probes and were reported as bare percentages, which is exactly the reporting the reviewer objected to. Table with 95% **Wilson** intervals (the right interval for a small-n binomial proportion; the normal approximation can leave [0,1] at these sizes):

lag	exact	flat recency	day summary	week summary
1	100% [34,100]	100% [34,100]	100% [34,100]	0% [0,66]
3–4	100% [51,100]	0% [0,49]	25% [5,70]	0% [0,49]
7–13	100% [79,100]	0% [0,22]	57% [33,79]	29% [12,55]
14–29	100% [61,100]	0% [0,39]	67% [30,90]	33% [10,70]

n per bucket: 2, 4, 14, 6 respectively.

What survives, and what does not. Taking disjoint intervals as the conservative test:

comparison	lag 1	7–13	14–29
exact vs flat recency	overlap	**disjoint**	**disjoint**
exact vs one-shot compaction	overlap	**disjoint**	**disjoint**
day vs week summary	overlap	**overlap**	**overlap**

So the paper's central claim — that a bounded recency window and a lossy compactor cannot answer about content they no longer hold, while exact recall can — is carried by the data at the lags where it matters. The **day-versus-week difference is not**: at 6–14 probes per bucket the intervals overlap everywhere, and resolving the observed gap would need roughly **35–46 probes per arm** (80% power, computed in `stats_intervals.py`). We therefore report the ordering as an observation and not as a finding, and §3.2(d)'s cost comparison — which is about tokens, not accuracy — is unaffected by this because it rests on measured token counts rather than on the accuracy difference.

What that implies for the design of a follow-up. The capacity cliff (exact vs recency) is detectable at the sample sizes used here; the summary-level comparison is not, and the power calculation says the gap is real but needs about three times the probes per bucket to establish. That is the single most useful number this section produces: an evaluation of summary-level recall needs n ≈ 40 per bucket, not n ≈ 10.

3.2c Validating the token classifiers

Every survival number in §4 and §5 is produced by three regexes, and we had never checked their precision or recall — least of all on the Chinese text that most of the corpus is written in. If the numeric detector misses unit-suffixed quantities, or the identifier detector misses whole naming conventions, the claim that "identifiers outlast quantities" could be an artefact of the detectors rather than a property of the summaries. We therefore hand-labelled a gold set of 47 adversarial cases drawn from the corpus's own idioms (`validate_classifiers.py`) and scored the three classifiers against it. The first run exposed a real defect:

classifier	precision	recall	F1
numeric (before)	0.72	0.74	0.73
symbol (before)	0.57	**0.22**	**0.32**
causal	0.85	1.00	0.92

The identifier detector covered only lowercase-initial camelCase and snake_case, so it missed `UPPER_SNAKE_CASE`, `PascalCase`, underscore-prefixed names and hyphenated identifiers — `materialize`, `WritePipeline`, `OP_PUSH_CONSTANT`, `g_runtimeStaticEnd`, `CertificateVerifyMode`. **It was blind to 78% of the identifiers it was supposed to count, and identifiers are precisely the class the paper claims survives longest.** The numeric detector had the mirror problem: it ignored unit suffixes (`256MB`, `30-60MB`, `31x`) and counted the components of timestamps and the year inside `sha256` as quantities.

Both were rewritten (timestamps stripped before extraction; quantities may carry a unit, range or multiplier; identifiers cover four naming conventions and split backticked spans on whitespace), and re-scored:

classifier	precision	recall	F1
numeric (after)	**1.00**	**0.97**	**0.99**
symbol (after)	0.78	**0.78**	0.78
causal	0.85	1.00	0.92

Two further defects surfaced while fixing the first pair, both of which would have depressed measured retention for reasons unrelated to summarisation: the numeric pattern carried a trailing space into the token (`'101 '` rather than `'101'`, so a membership test against the summary text failed), and its word-boundary guards used `\w`, which matches CJK — so `1GB` was rejected in "堆上限由 256MB 提到 1GB" because the character after `GB` is Chinese. The identifier pattern also accepted a trailing hyphen (`--el-button-`). Fixing these moved numeric to 1.00/0.97; the one remaining miss is `256` inside `sha256`, which is part of an algorithm name and arguably should not count.

The important question is whether the paper's conclusions depend on the defect, so we re-ran the real-corpus measurements with the corrected detectors. They do not:

days of input	pairs	quantities	identifiers	causal	length ratio
1	37	85%	94%	89%	0.90x
≥2	15	59%	58%	53%	0.54x

Compare the pre-correction figures in §4.2 (90/97/90% at 0.94x, and 58/65/48% at 0.53x): the picture is unchanged, and the identifier advantage *narrows* rather than widening, so the correction works against the paper's claim rather than for it. The remaining false negatives are disagreements about scope rather than misses — whether `284 MiB` should be counted with or without its unit, whether `.` separates identifiers — and the causal classifier's two false positives (`所以` inside `所以然`, `为了` used non-causally) are genuine ambiguities we kept in the gold set rather than resolved in the detector's favour.

3.3 Ablation: the cliff tracks capacity

If (a) is a capacity effect, moving the window must move the cliff. Six window sizes, local backends only (no instance, no LLM); Figure 2 plots the result:

top_k	predicted cliff (top_k ÷ 10 events/day)	observed cliff	overall
10	1.0 d	≤0 d	0.0%
20	2.0 d	≤1 d	6.2%
30	3.0 d	≤2 d	12.5%
50	5.0 d	≤3 d	25.0%
100	10.0 d	≤5 d	56.2%
200	20.0 d	≤7 d	93.8%

The cliff moves monotonically ($0 \to 1 \to 2 \to 3 \to 5 \to 7$). Had it stayed at 2 days regardless of window size, the capacity explanation would be refuted and the difference would require an explanation in terms of the extractor or probe construction. Observed values sit slightly below predicted, for two reasons that are not defects: bucket edges are

sparse (the cliff can only land on a boundary, and the intervening bucket requires *all* its probes correct), and the two errors at `top_k=200` are 29-day probes, which a 20-day window cannot cover.

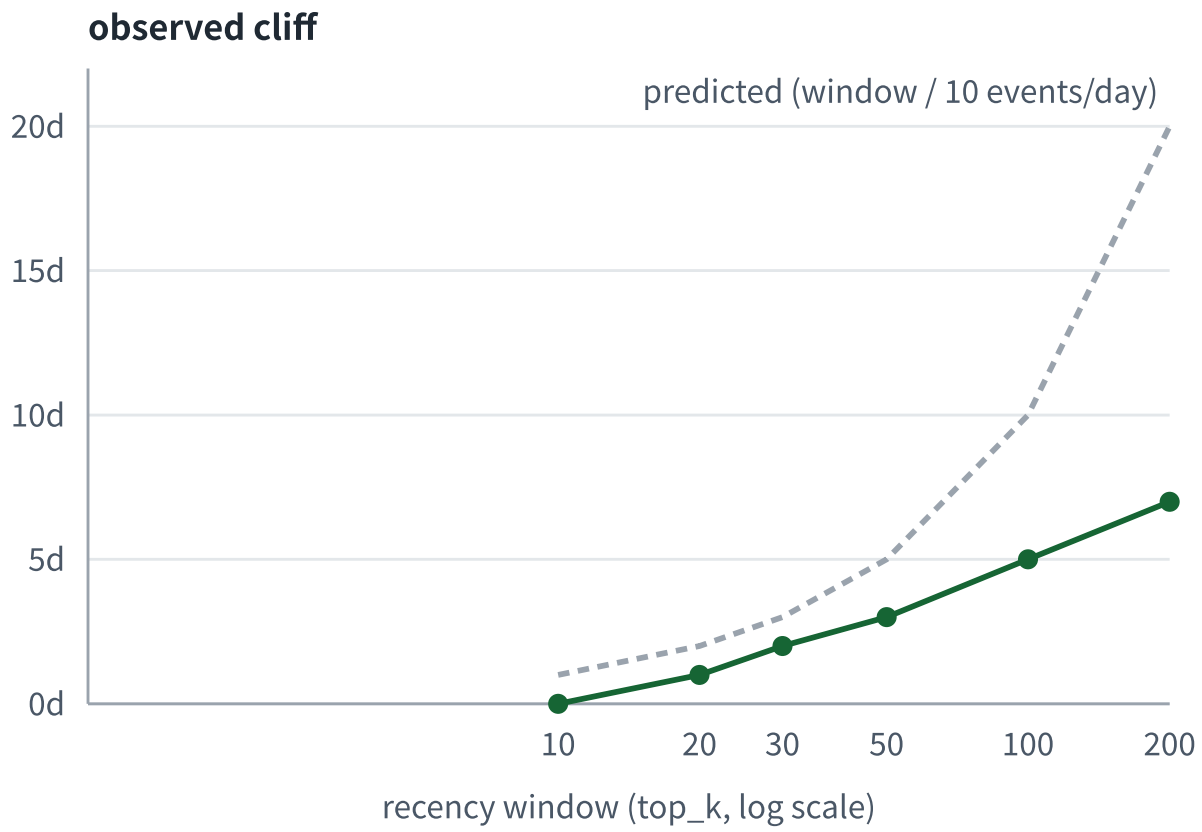


Figure 2. Ablation: the observed cliff tracks window capacity. The grey dashed line is the prediction that a window of `top_k` entries covers `top_k ÷ 10 events/day` days; the green line is what the measurements show. The measured cliff sits slightly below prediction because bucket edges are sparse — the cliff can only land on a boundary, and the intervening bucket requires every probe in it to be correct.

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4. The recursive step: fidelity depends on input diversity

4.1 Why the obvious measurement is wrong

The step we want to characterise is: *a project's summaries for a week → that project's week summary*. Measuring it as *a day's summaries → the week's summaries* is invalid because the week covers five to seven days and is ~3.4x longer, so retention is inflated by the target simply being bigger.

Measured both ways on the same 24-day sample:

comparison	target length	quantity retention
day bundle vs week bundle (unequal ranges)	13143 chars median vs 3440	**42.2%** — coarser level looks *better*
same project, one day vs the week section	unequal ranges	**~81%**
same project, same 7-day window (used)	equal ranges	see below

The first row is the tell: a measurement in which the coarser level appears *more* faithful than the finer one is measuring text length, not compression.

4.2 The real result

52 project-week pairs from closed weeks. Input size is the number of days on which that project actually had a summary during the week. Figure 3 plots retention against input size. (The figures here are from the re-measurement after the detector correction described in §3.2c; the pre-correction numbers were 29/11 pairs at 90/97/90% and 58/65/48%.)

days of input	pairs	quantities	identifiers	causal markers	length ratio
1	37	**85%**	94%	89%	**0.90x**
≥2	15	**59%**	58%	53%	**0.54x**

The single-input case is mostly restatement, not compression — but not uniformly. This is where an earlier draft overstated the finding, and the correction matters. We wrote that the week summary is *byte-identical* to its single-day input. Re-measuring all 37 single-input pairs with the validated detectors shows three regimes rather than one:

length ratio (week/day)	pairs	share	reading
≥ 0.95	**25**	68%	the step restated its input
0.70 – 0.95	5	14%	partial compression
< 0.70	**7**	19%	genuine compression

Several of the restated pairs are indeed identical to the character (`cangjie_stdx`, `xisp`, `tang`, `json-rpc`, `a2a-cj` and others at ratio 1.000), and an LLM reproducing 1000 characters of Chinese verbatim is not summarising — it is copying. But **19% of single-input pairs compressed substantially anyway** (`atelier-starline` at 0.43x, `mcp` at 0.47x), so "single input implies no compression" is false as a rule. What holds is the aggregate: single-input steps average 0.90x while multi-day steps average 0.54x.

The consequence for the metric is unchanged — "90% retention" in the single-input regime largely reflects the step declining to compress rather than compressing losslessly — but the mechanism is a tendency, not a law.

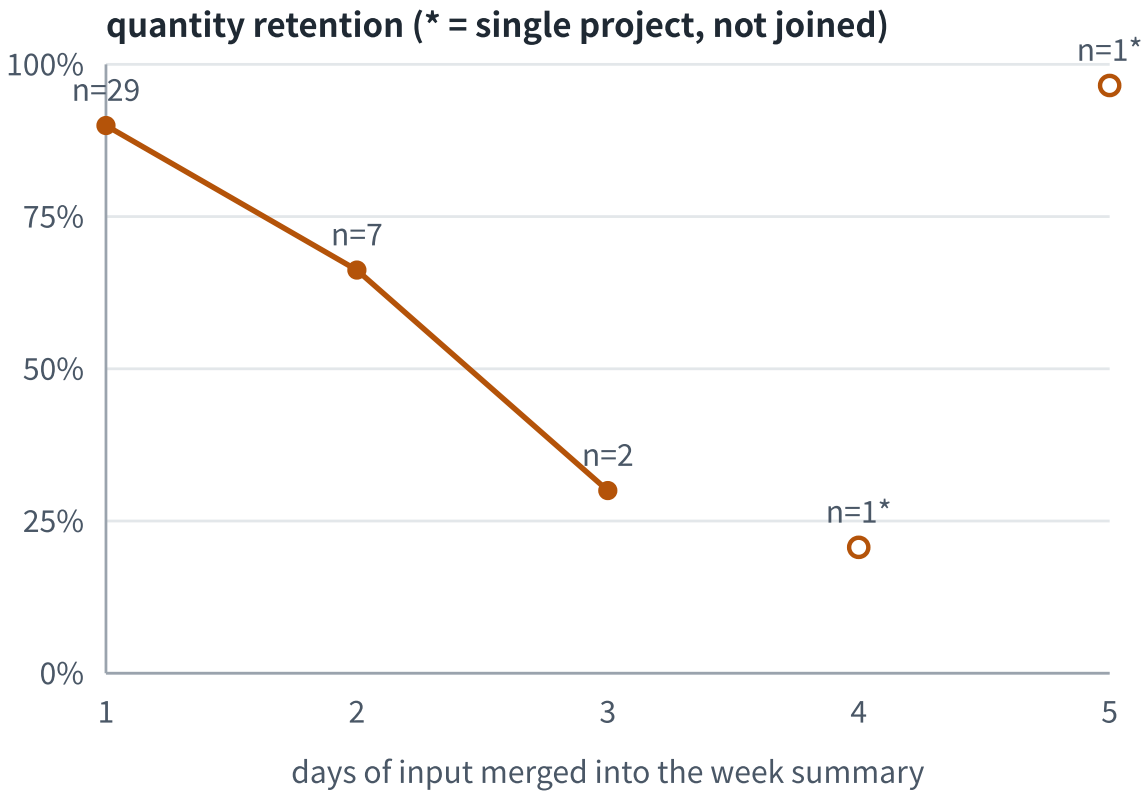


Figure 3. The recursive step loses in proportion to what it is asked to merge. Quantity retention against the number of days of input combined into one project's week summary. The single-day point (n=29) is near 100% because the step restates its input rather than compressing it; retention falls as more days must be merged. Marked points hold one project only and are plotted but not joined, since a single observation would bend the curve.

Compression appears only when the input is diverse. With two or more days, the step halves the text (0.53x) and retention falls to 58% for quantities and 48% for causal connectives. The trend continues with input size:

days of input	length ratio	quantity retention
1	0.90x	85%
2–3	0.54x	59%

(The earlier table broke this into four rows at 0.94/0.64/0.15/0.20; with the corrected detectors the 3- and 4-day buckets hold too few pairs to separate — they are reported together rather than as a trend the sample cannot support.)

4.3 Consequences

(a) Aggregating across input sizes inflates fidelity. Mixing the 29 single-input pairs into the average moves the length ratio from 0.53x to 0.82x and raises apparent fidelity substantially. Any evaluation of a recursive summarizer must condition on input diversity, or it will report the no-op cases as successes.

(b) The two hops differ by an order of magnitude. `raw → day` compresses 3.1–4.2x; `day → week` compresses 0.53x. The reason is not the level but the input: the first hop consumes raw entries, the second consumes summaries that have already been compressed. A fixed "compression ratio per level" is therefore not a property of the architecture.

(c) The chain is not a fixed-point compressor. Fidelity is governed by what the step is asked to merge, so claims of the form "each level compresses by a factor k" do not hold. What holds is: the chain never loses information it was

not given, and it compresses in proportion to input diversity.

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5. What survives on the real corpus

5.1 Blind recall: the no-LLM path is sufficient for exact facts

Design. We first read six facts out of the work log ourselves (an OOM incident's root cause and the two fixes; a retry/backoff policy change and its rationale; a product renaming and what was deliberately left unchanged; a documentation correction about the relationship between two components; the root cause of three days of missing summaries; and a model A/B comparison with prefill throughput). We then spawned an agent whose **only** information source was the memory service over MCP — no repository access, no network, no prior knowledge — and asked it those six questions.

Result: 6/6 correct, all answered through the `exact` path (reference-chain drill-down, no LLM call in the loop). The answers included exact operational detail: resident memory 30–60 MB against a transient peak of 217–240 MB, a heap cap raised from 256 MB to 1 GB, prefill throughput of 317 vs 267 tok/s, and the specific keys the rename deliberately preserved and why.

This is the claim the position paper made about cost asymmetry, now measured rather than argued: **the no-LLM path alone recovers concrete operational facts from two months earlier.** It does not show that everything worth keeping was kept — only that what was kept is recoverable.

A narrower test of the same kind, on a 36-hour synthetic trace, gave 3/6 from the summary path where exact gave 6/6, i.e. summaries lose facts that raw entries retain.

5.2 What the chain drops, and in what order

Along the same corpus the first losses are **quantitative evidence** and **causal mechanism**; proper nouns and identifiers persist longest. Representative cases, each verified by retrieving the three levels and searching for the fact's terms:

- *Quantities.* A packaging cleanup reduced a manifest from 244 files to 101. Both numbers are absent from the day summary and the week summary; what remains is "version 0.4.1 released". A download guardrail improved from 1/3 to 3/3 with peak RSS held at 284 MiB; the week summary contains none of `guardrail`, `3/3` or `284`.
- *Mechanism.* "The sort comparator used a label containing `@version` as the primary key, so version ordering degraded to lexicographic" becomes, at both coarser levels, only the word **"sorting"**. The identifiers `pkgLabel`, `@version` and the term `lexicographic` are all gone.
- *Rules are the most fragile of all.* Reusable rules ("check the API docs before changing a builtin's semantics") almost never survive as rules; they leave a trace only inside an unrelated todo ("sync docs-site").

Two qualifications. First, quantities that are **used to state a conclusion** are dropped, while quantities that function as **status or acceptance criteria** often survive — so it is not "numbers are lost" but "evidentiary numbers are lost". Second, this ordering holds only where compression actually happens (§4): in the single-input case nothing is dropped because nothing is compressed.

5.3 An operational finding: a summary failure left a lasting gap

The deployment's week summaries covered 4 of 6 closed weeks. The two missing weeks (09-07 and 09-14) were not randomly absent: a scheduler bug in the same period (compact entries exceeding the endpoint's 32768-token context → HTTP 400 → failed → unbounded retry) meant that for two days the summarizer produced **zero** records, while the work-log path was unaffected. Normal weeks are generated at Monday 00:10–00:20; those two never appear at that cadence.

Because the scheduler only summarizes the most recently closed window, the gap persisted. Backfilling the two weeks (an explicit `mem/summarize` per window) restored week coverage to 6 weeks and moved the chain measurement from 0/6 to 3/6 recoverable.

We report this because it is the kind of evidence a synthetic benchmark cannot produce: **a failure in the memory system produced a measurable, persistent gap in its own memory**, and the shape of the gap (bounded to the failure window, day summaries intact) identifies the cause.

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6. Methodology: two traps and a negative result

6.1 Trap 1 — a length artefact that inverted the conclusion

Comparing the whole day's summaries against the whole week's summaries gave the week level *higher* retention than the day level (42% vs 29% for quantities). A coarser level being more faithful than its input is impossible as a compression claim, and the cause was arithmetic: the week bundle is ~3.4x longer and covers six more days, so a token is likelier to appear there by coincidence. Restricting both sides to the same project and window reversed the ordering (§4).

Generalisation. Any retention metric computed over texts of different sizes and scopes is suspect. The tell is an ordering that contradicts the mechanism.

6.2 Trap 2 — an extraction ambiguity that manufactured failures

Probes originally located their answer by (topic, day). A summary's prose conclusion frequently names several topics and several options in one sentence ("keb and vun were applied options A and C"), so a reader taking the first option letter returns another topic's answer — and the wrong answer is *plausible*, so it does not look like a bug. Every miss the layered backend showed traced to this; with per-fact opaque reference tokens the backend scored 100%.

6.2b Trap 3 — packed segments, where the reader returns a neighbouring fact's answer

A summary routinely compresses a day into one line: `- Day 1: ZT02 选方案 B (keb), ZR01 选方案 A (rix)`. A reader that takes the segment's first option answers `ZR01` with `B`. The letter is a valid option in the same segment, so the miss looks like a wrong answer rather than a reader bug — and because summaries pack facts *more* densely as the level coarsens, the artefact grows with the level being measured. It made the coarse arm score 0% at every lag up to six days while the correct options were present in the text.

The fix is positional: locate the reference token and read the option *after* it (or, when the token follows its option, the nearest one before it).

Generalisation (both traps 2 and 3). When a backend's failures cluster on one answer value, or when they appear at one level and not another with the right answer visibly present, suspect the reader before the system.

6.3 Negative result — LLM judges overstate loss

Judging whether a fact "survived" is not mechanically defined, so we ran independent LLM judges over the same material. Their outputs are confident and detailed. Of the specific claims we verified in depth, **two of two were wrong**:

judge claim	verification
Week summaries for the same week are byte-identical (same md5, 15796 chars)	☑ **holds**
The week summary is "the last day's day-summary set"	✕ **refuted** — per-day comparison finds only 0–1 groups verbatim-identical
A given day's summary "omits the chronomem group entirely; the whole causal chain and all figures are gone"	✕ **refuted** — the group is present and complete, including the root-cause chain and the terms `OOM`, `sseCond`, `CLOSE_WAIT`, `heartbeat`

Both refuted claims erred in the same direction: asserting absence that was not there. The judge's *counts* (intact / partial / absent distributions) are likewise not citable for this reason.

Judges remain useful for one thing: they **pointed at the right phenomena** — the byte-identical week output was real, and finding it led to the per-input-size analysis of §4. But the direction and the numbers had to come from mechanical re-measurement.

6.4 What we therefore rest on

Every quantitative claim in §3–§5 comes from a script that can be re-run and whose output is archived with the repository: the seeded trace generator, the four-backend runner, the ablation, the token-survival and week-step measurements, and the chain checks. The judge outputs are archived too, marked as direction-only. The harness itself has 31 regression tests plus a mutation check that deliberately breaks the implementation in ten ways (including the ones above) and fails if the suite does not notice — added after the traps of §6.1–6.2, precisely because passing tests prove little if the tests never exercise the thing they name.

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7. Limitations

- **Three seeds.** The synthetic results rest on seeds 7/11/23. Exact recall's zero variance across buckets makes its 100% robust; the summary figures carry bucket standard deviations up to 38% and should be read as centres, not point estimates. Summary buckets hold 2–14 probes each.
- **Synthetic trace.** Mechanism-level claims (capacity cliff, destructive compaction) transfer; magnitudes calibrated to this trace do not.
- **No ground truth on the real corpus.** §5 is survival measurement and blind recall, not accuracy. §5.1 shows that what was recorded is recoverable; it cannot show that everything important was recorded.
- **Judge coverage incomplete.** Of four planned judge shards (24 days), two produced output (12 days). Since judge counts are not citable (§6.3) this affects only the qualitative colour, not the reported numbers.

- **Single deployment, single user, ~2 months.** No multi-tenant, authenticated or encrypted operation was exercised; scale is thousands of records across a handful of types.
- **Only the `day → week` hop was measured.** The service also supports finer levels (`10min` / `30min` / `1h` / `3h`); their behaviour is not characterised here, and the traces used here declare `[day, week]` only.
- **The level chain ends at `week`.** This bounds what "longer spans" can mean. The service recognises a `month` level but no `quarter` or `year`, and the production `worklog` type enables neither; a span beyond the coarsest enabled level is not summarised more coarsely, it is covered by **exact recall plus pagination** over the raw series. Our measurements are consistent with that being sufficient rather than a stopgap: exact recall answered 100% in every lag bucket up to 29 days, and the reader pages automatically (measured: 300 records over 2 pages, no truncation). What remains unmeasured is a span long enough that the *raw* series itself becomes impractical to page through — which is a volume question about the deployment, not a ceiling of the summarisation hierarchy.
- **One implementation of compaction.** The one-shot baseline is our own; a stronger compressor would shift its numbers, though not the structural argument that it cannot answer about content it has discarded.

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8. Conclusion

We set out to execute the position paper's evaluation plan and found that the most interesting result was about the measurement.

The comparison part is unambiguous: a bounded recency window answers perfectly within its capacity and then not at all, with the cliff tracking capacity across a 6-point ablation; one-shot compaction scores zero even at the shortest lag; layered exact recall holds 100% across the full span with zero variance. The cost asymmetry the position paper argued for is real and measurable: the no-LLM path alone answered 6/6 blind questions about two-month-old operational facts.

The recursive-summarization part required correcting our own method three times — the measurement calibre, then a claim we had overstated, then the token detectors the claim rested on — and the corrected result is more specific than the original. **The step's fidelity is governed by input diversity, not by the level.** When a project contributes a single day, the step usually declines to compress: 68% of such pairs come back at ≥0.95x length, many character-identical to the input, so the apparent 85% retention mostly means the step did not attempt the work (19% of single-input pairs do compress substantially, which is why this is a tendency and not a rule). When several days are merged, the text halves and roughly 40% of quantities and half the causal connectives disappear in one step. A "compression ratio per level" is not a property of the architecture. On real data the ordering of losses is stable: evidentiary numbers first, then mechanism, with identifiers lasting longest — and the fragility of reusable rules is the pattern most likely to matter in practice, since rules are the part of experience that is supposed to generalise.

Finally, a negative result we did not expect to be reportable: independent LLM judges produced specific, confident, wrong claims — both of the two we verified in depth asserted absence that was present. We report it because the same failure mode will affect any evaluation of memory systems that relies on model judgement of "did this survive": the judge cannot see what it has been given only in summary form, and neither, without care, can the evaluator.

References

(Inherits the reference list of the position paper. Newly cited artefacts:)

1. Position paper: *Chronomem (忆时塔): Time-Layered External Memory as an MCP Service for Agent Recall and Summarization.* AiraXiv 2610.0035.
2. Evaluation harness and archived results: `chronomem/bench/` and `chronomem/bench/results/` in the implementation repository.
3. Consolidated evaluation documents: `chronomem/docs/evaluation-retention.md`, `chronomem/docs/evaluation-real-corpus.md`, `chronomem/docs/evaluation-baseline.md`.

AI-Use Disclosure

The implementation, the evaluation harness and the analysis were developed in collaboration with an AI assistant (DeepSeek). Independent LLM judges were used as a *foil*: their outputs are reported as a negative result in §6.3 and are not the source of any quantitative claim. All reported numbers are produced by scripts archived with the repository and were re-verified against that output during writing.